

L6 Functions

For Discord email - support@learnyard.com

What are Functions ?

Sum of 2 number → Print

I → access
specifier

II → static

public static return type functionName (parameter)

{

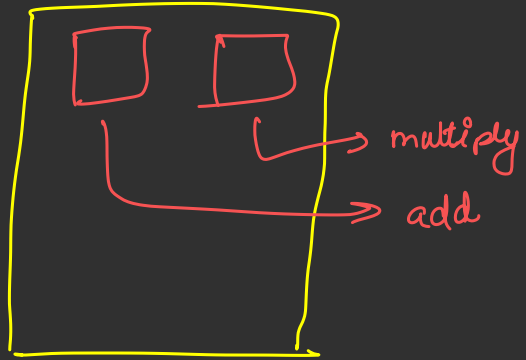
// Logic Part

}

input

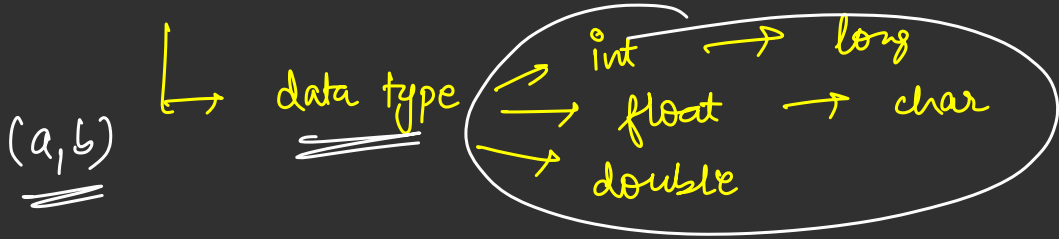
Direct usage of
Functionality
↓
Static

Class →
↓
Blueprint



non-static Functionality → object
↙
Indirect
↘
Key →

I → Calculation & Final result



II → logic / Print }

Algo

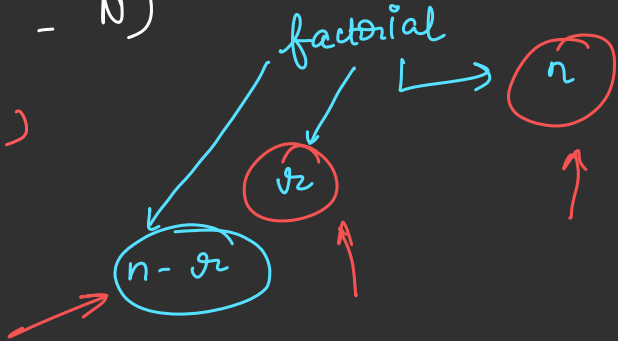
void

Use-case of Functions

Armstrong

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

```
for ( 1 - - - - N )  
{  
    if ( Armstrong )  
}
```



$${}^nC_r = \frac{n!}{r! (n-r)!}$$

$$\checkmark 1 \dots n$$

↑

$n!$

5! → factorial

$$1 * 2 * 3 * 4 * 5$$

$$1 \dots n$$

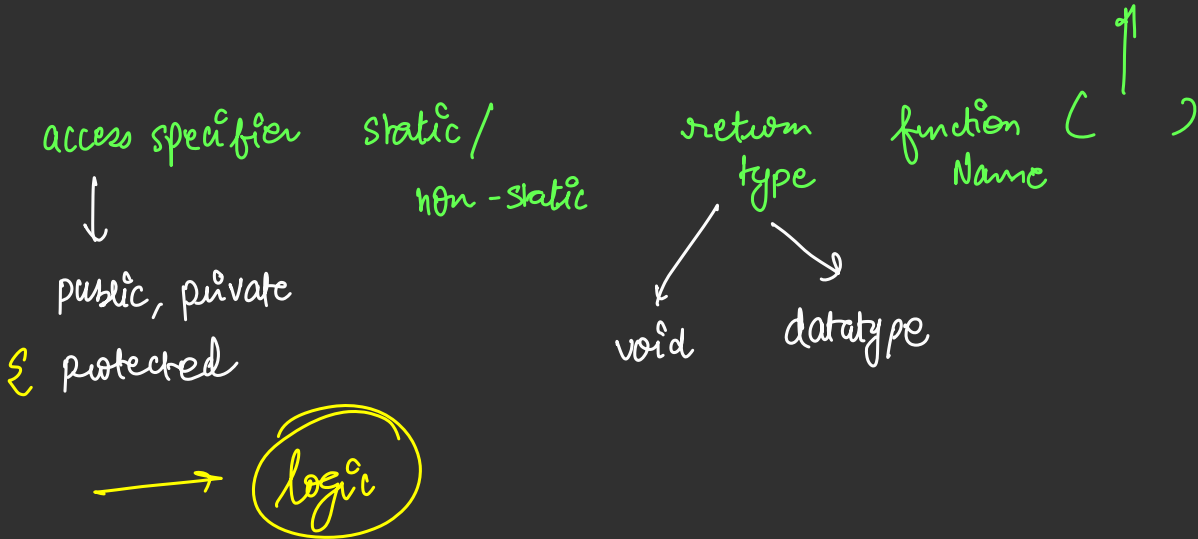
$$\checkmark r! \rightarrow 1 \dots r$$

$$\checkmark (n-r)! \rightarrow 1 \dots (n-r)$$

$$\text{factorial}(n) \rightarrow n! \rightarrow \textcircled{1}$$

Different Aspect of Functions

Parameter



Passing Arguments in Function

```
public class Main  
{
```

```
    public static int sum ( int a, int b, int c )
```

```
    {  
        int result = ( a + b + c );
```

```
        return result;
```

```
    public static void main ( String arg [] )
```

```
    {  
        int x = 5 ; int y = 6 ; int z = 7 ;
```

```
        int s = sum ( x, y, z );
```

Execution

5 6 7

↑ ↑ ↑

Print
↑
Algo
Logic

void

Sum of number using Functions

```
public class Main
```

```
{  
    public static
```

void

sum (int a, int b)

```
{  
    s.o. pln (a + b);
```

19

```
}  
public static void main (String args[])
```

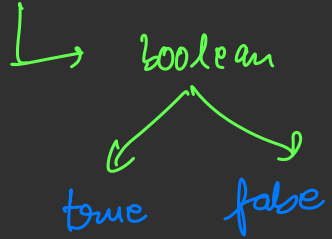
```
{  
    sum (10, 9);
```

Print
10 9
↑ ↑

Even odd number using Functions

Odd $\rightarrow$ Not divisible by 2

Even $\rightarrow$ Divisible by 2



```
public static boolean even(int n)
```

```
{  
    if (n % 2 == 0)  $\rightarrow$  even number
```

```
    {  
        return true;  
    }
```

```
    else
```

```
    {  
        return false;  
    }
```

} $\rightarrow$ 1 + return type

N = 6

Understand how to submit questions on Leetcode

We have to complete functions on leetcode

Scope of Variables in Function

Pass by Value and Pass by Reference in Function

Call stack working in Function

Can we write function inside another function ?

Extra question [Armstrong using Function]

Practice and Homework Questions

Homework questions -

<https://codeforwin.org/c-programming/c-program-to-find-maximum-and-minimum-using-functions>

<https://codeforwin.org/c-programming/c-program-to-find-prime-numbers-between-two-intervals-using-function>

<https://codeforwin.org/c-programming/c-program-to-print-armstrong-numbers-between-1-to-n-using-function>

Thank You!

Please practice more questions and examples as above !!